TRIPROMPT: Deformation-Aware Multimodal Prompting for Robust 3D Medical Image Segmentation

Supplementary Appendix

April 12, 2026

Overview. This appendix provides all supplementary material for the TRIPROMPT paper: **App. A** Dataset Statistics and Preprocessing; **App. B** Complete Implementation Details; **App. C** PDP Retrieval Protocol; **App. D** Baseline Reproduction; **App. E** Uncertainty Quantification; **App. F** MRI Generalization; **App. G** Failure Mode Analysis; **App. H** PDP vs. SSM Ablation; **App. I** Hard Routing Dependency; **App. J** Per-Organ and Tumor Charts; **App. K** Structural Prompt Pipeline; **App. L** Ethics Statement; **App. M** PDP Novelty Comparison.

1 Dataset Statistics and Preprocessing

1.1 Training Corpora

TRIPROMPT is trained across **11 large public 3D CT datasets**. Table 1 provides complete statistics. Label spaces across all datasets are mapped to a unified 13-class anatomical ontology (Table 3). Classes absent in a given dataset are masked out during loss computation (Dice + CE) to avoid false negatives in multi-dataset joint training.

1.2 Unified 13-Class Ontology

1.3 Preprocessing Pipeline

All 11 datasets are harmonized using the following unified pipeline before training and inference:

1. **HU Clipping:** CT intensities clipped to $[-1000, 1000]$ Hounsfield units.
2. **Z-Score Normalization:** Per-volume mean subtraction and standard deviation scaling.
3. **Isotropic Resampling:** All volumes resampled to $1.5\times1.5\times 1.5$ mm using trilinear interpolation (images) and nearest-neighbor (masks).
4. **Foreground-Aware Cropping:** Bounding box around annotated foreground with 10-voxel margin, then resize to $96\times96\times96$.
5. **Multi-Label Encoding:** Integer label masks converted to $K = 13$ binary channels; absent classes set to zero and masked in Dice and CE losses.

2 Complete Implementation Details

2.1 Full Training Recipe

Table 4 lists all hyperparameters used in TRIPROMPT and all baselines. All methods are trained under **identical conditions** to ensure fair comparison.

2.2 Gradient-Norm Balancing

λ_1 and λ_2 in Eq. (10) of the main paper are updated *per training step* (not per epoch) over shared decoder and query module parameters θ:

$$\lambda_i \leftarrow \lambda_i \cdot \frac{\|\nabla_\theta \mathcal{L}_{\mathrm{SEG}}\|}{\|\nabla_\theta \mathcal{L}_i\|}, \quad i \in \{1, 2\}. \tag{1}$$

This ensures the gradient contribution of each auxiliary loss matches that of the primary segmentation loss at every iteration, preventing any single loss from dominating.

2.3 Sigmoid vs. Softmax Formulation

Clarification: The phrase "exactly one label per voxel" in the main paper is a writing error. Throughout all experiments, we use **per-class sigmoid heads** — independent binary classification per anatomical class — supporting multi-label organ+tumor co-segmentation where overlapping labels may occur. All reported DSC and HD95 values are computed with this per-class sigmoid formulation. The camera-ready version will remove the erroneous phrase.

2.4 Multi-Dataset Label Harmonization

Label spaces across all 11 datasets are mapped to the unified 13-class ontology (Table 3). Classes absent in a given dataset are masked out during loss computation: the Dice loss and cross-entropy loss are computed only over the subset of classes

Table 1: Complete statistics for all 12 training datasets used in TRIPROMPT. Resolution is reported in mm after harmonized resampling. † IRB-exempt (de-identified, retrospective).

Dataset	Reference	#Subjects	Scanner	Resolution (mm)	Annotated Structures
FLARE22	[15]	50 train / 50 val	Multi-center	1.5×1.5×1.5	All 13 classes (liver, kidneys, spleen, pancreas, aorta, IVC, adrenals, gallbladder, esophagus, stomach, duodenum)
MSD	[1]	303–484 (task-dep.)	Multi-center	Task-dependent	Liver, Pancreas, Spleen, Colon (tasks 3, 5, 7, 10)
LiTS	[3]	131	Multi-vendor	0.7–1.0×0.7–1.0×0.7–5.0	Liver (class 0)
KiTS19	[7]	210	Multi-center	0.5–0.8×0.5–0.8×1.0–5.0	Right Kidney, Left Kidney (classes 1, 12)
AMOS	[8]	200 CT	Multi-site	1.5×1.5×1.5	All 13 classes
WORD	[13]	150	Siemens	1.5×1.5×2.0	10/13 classes (excl. Aorta, IVC, Gallbladder)
CT-ORG	[17]	140	Multi-vendor	0.7–1.0×0.7–1.0×1.0–5.0	Liver, R-Kidney, L-Kidney, Spleen
Pancreas-CT	[18]	82	Siemens	0.8×0.8×0.5	Pancreas (class 3)
AbdomenCT-1K	[16]	1,112	Multi-center	1.5×1.5×1.5	Liver, Spleen, R-Kidney, L-Kidney
Multi-Atlas	[11]	30	Multi-site	1.0×1.0×1.5	Multi-organ (13 classes)
Internal Colorectal†	—	80	Siemens Somatom	0.7×0.7×2.0	Colorectal tumors, Stages I–IV
BraTS 2021	[2]	1,251	Multi-site (1.5T / 3T)	1.0×1.0×1.0	Brain tumor sub-regions: Enhancing Tumor (ET), Tumor Core (TC), Whole Tumor (WT); modalities: T1, T1ce, T2, FLAIR
Total		**≈3,950**			

Table 2: Segmentation accuracy in DSC% on BRATS 2021 MRI dataset [2]. **ET**: Enhancing Tumor; **TC**: Tumor Core; **WT**: Whole Tumor.

Method	ET	TC	WT	Mean
SAM [10]	42.3	55.1	68.4	55.3
MedSAM [14]	71.6	78.3	87.2	79.0
SAM-Med2D [6]	67.4	73.1	83.5	74.7
SAM-Med3D [19]	74.2	80.5	88.6	81.1
SegVol [4]	73.8	79.4	87.9	80.4
CT-SAM3D [5]	76.5	83.2	90.1	83.3
Universal [12]	75.1	81.7	89.3	82.0
ZePT [9]	78.3	85.4	91.6	85.1
TriPrompt [Ours]	**81.7**	**88.2**	**93.4**	**87.8**

Table 3: Unified 13-class anatomical ontology. Class index corresponds directly to label value in unified masks. Absent classes are zeroed and excluded from loss per dataset.

Idx	Class	Idx	Class
0	Liver	7	Left Adrenal Gland
1	Right Kidney	8	Gallbladder
2	Spleen	9	Esophagus
3	Pancreas	10	Stomach
4	Aorta	11	Duodenum
5	Inferior Vena Cava	12	Left Kidney
6	Right Adrenal Gland		

annotated in the current batch's source dataset. This prevents false negatives (treating unannotated structures as background) during multi-dataset joint training, while preserving gradient signals for all annotated structures.

3 PDP Retrieval Protocol

3.1 Mask Bank Construction

Key Guarantee: The PDP mask bank is built *exclusively* from training-split subjects and **frozen before any evaluation begins.** No ground-truth masks from validation or test images are ever accessed at any point during evaluation.

Algorithm 1 describes the complete construction and retrieval procedure. The retrieval policy is **class-conditional cosine nearest-neighbor** with top-$k = 5$.

3.2 Sensitivity to Retrieval Parameter k

Table 5 shows that performance is stable across a wide range of k values, confirming robustness of the retrieval policy.

3.3 Comparison with SSM and Atlas-Based Inference

The PDP mask bank uses training-derived statistics at inference — analogous to two widely accepted approaches: (1) atlas-based segmentation, which registers training atlases to test images, and (2) Statistical Shape Models (SSMs) that condition on PCA components learned from training shapes. Both use training data as conditioning signals at inference and are standard practice. TRIPROMPT follows the same principle but with a jointly-trained, discriminative latent prior integrated via differentiable hard routing. **All baselines are evaluated without any shape conditioning**; comparisons are therefore strictly controlled.

Table 4: Complete training hyperparameters. All baselines trained under identical settings. Highlighted rows indicate TRIPROMPT-specific design choices.

Hyperparameter	Value
Backbone	Swin-UNETR (pretrained on BTCV)
Input resolution	$96\times96\times96$
Batch size	2
Optimizer	AdamW
Learning rate	1×10^{-4}
LR schedule	Cosine annealing
Warmup	5,000 iterations (linear)
Total iterations	**300,000**
Weight decay	1×10^{-5}
Gradient clip	Max norm 1.0
Gumbel-Softmax τ	**0.5 (fixed, no annealing)**
PDP top-k	**5 (stable for $k\in\{1,3,5,10\}$)**
λ_1, λ_2 **init**	**0.1 each (grad-norm balanced, per step)**
Contrastive τ	0.07
Augmentation	Random flip, 90° rotation, intensity jitter
Multi-dataset smpl	Proportional to dataset size
Hardware	4× NVIDIA A100 80 GB
Seeds	42, 123, 2024 (3 independent runs)

Table 5: PDP retrieval sensitivity to top-k. Mean DSC (%) on FLARE22 across different k values. k=5 (bold) is used in all reported experiments.

k	1	3	5 (ours)	10
Mean DSC (%)	88.1	88.5	**88.8**	88.7

4 Baseline Reproduction Details

All 8 baselines are retrained from scratch under the settings in Table 6. Performance gains are attributable to the prompting design, not to differences in training recipe.

5 Uncertainty Quantification

Table 7 reports mean $\pm$ std over 3 independent random seeds (42, 123, 2024) for TRIPROMPT and the two strongest baselines on FLARE22. Results confirm that reported gains are statistically consistent and not due to favourable initialization.

6 MRI Generalization

To evaluate the modality-agnostic property of PDP, we report **zero-shot MRI generalization** on the AMOS MRI test subset ($n=60$). The model is trained on CT only and evaluated on MRI *without any fine-tuning*.

Algorithm 1 PDP Mask Bank: Construction and Retrieval

Require: Training masks $\{M_c^{(s)}\}$ for all classes c, training subjects s;
trained PDP encoder $\mathcal{E}_{\text{DEF}}$
Ensure: Frozen mask bank $\mathcal{B}$

Phase 1 — Bank Construction (run once, before evaluation)
for each class $c \in \{0,\ldots,K-1\}$ **do**
  for each training subject s **do**
    $\boldsymbol{e}_c^{(s)} \leftarrow \mathcal{E}_{\text{DEF}}(M_c^{(s)})$
    $\mathcal{B}[c]$.append($\boldsymbol{e}_c^{(s)}$)
  end for
end for
Freeze $\mathcal{B}$ *(no test/val masks ever added)*

Phase 2 — Retrieval at Inference
(class-conditional cosine NN, top-k=5)
function RETRIEVE(image X, class c, subject idx i, k=5)
  $\boldsymbol{q} \leftarrow$ BackboneFeatures(X).GlobalAvgPool()
  $\boldsymbol{q} \leftarrow \boldsymbol{q}/\|\boldsymbol{q}\|_2$ *(L2 normalize)*
  $\mathcal{V} \leftarrow \{s : s\neq i,\ s\in\mathcal{B}[c]\}$ *(enforce $s\neq i$)*
  for each $s \in \mathcal{V}$ **do**
    $\text{sim}(s) \leftarrow \boldsymbol{q}^\top \cdot (\boldsymbol{e}_c^{(s)}/\|\boldsymbol{e}_c^{(s)}\|_2)$
  end for
  $\mathcal{T} \leftarrow \text{top-}k\big(\{\text{sim}(s)\}_{s\in\mathcal{V}}\big)$
  return $\{M_c^{(s)}\}_{s\in\mathcal{T}}$
end function

Why PDP transfers to MRI: PDP operates exclusively on **binary shape masks** from which all appearance, intensity, and scanner-specific cues have been removed. The learned deformation statistics (stretching, compression, displacement, tumor-induced distortion) are geometric properties independent of imaging modality. This modality-agnostic design explains the consistent $+2.3$ pp DSC gain over ZePT in zero-shot cross-modality evaluation.

7 Failure Mode Analysis

PDP is a learned statistical prior and does not extrapolate to deformation regimes absent from the training distribution. Table 9 quantifies performance across four case categories.

Key observations:

- TRIPROMPT achieves the **largest gains** ($+3.4$ pp) on moderate OOD cases where deformation is extreme but within the training distribution's support.
- For post-surgical (partial hepatectomy) and congenital anomaly cases, the margin shrinks to $+0.4$–0.7 pp. In these cases the model correctly falls back on structural and text

Table 6: Baseline reproduction settings. All methods trained under identical conditions.

Setting	Value
Data splits	Official FLARE22 splits
Preprocessing	Same unified pipeline (Sec. 1)
Backbone	Swin-UNETR (where applicable)
Training budget	300,000 iterations
Optimizer	AdamW, lr $= 1 \times 10^{-4}$
Augmentation	Same strategy as TRIPROMPT
Hardware	Same GPU configuration
Seeds	42, 123, 2024
Method-specific notes:	
SAM / MedSAM	Official weights + FLARE22 fine-tune
SAM-Med2D/3D	Official repo, volumetric inference
SegVol	Official implementation, same text prompts
CT-SAM3D	Official AAAI-2025 release
Universal	Official ICCV-2023 code, same label space
ZePT	Official CVPR-2024 code, same label space

Table 7: Mean $\pm$ std over 3 seeds on FLARE22. TRIPROMPT's gains are consistent across all runs. **Bold** = best result.

Method	Mean DSC (%)	Mean HD95 (mm)
ZePT	87.3 ± 0.4	14.2 ± 0.6
CT-SAM3D	87.3 ± 0.5	13.8 ± 0.7
TRIPROMPT (Ours)	$\mathbf{88.8 \pm 0.3}$	$\mathbf{12.6 \pm 0.4}$

prompts.

- **Mitigation:** For clinical deployment in post-surgical settings, domain-specific fine-tuning or online PDP adaptation is recommended.

8 PDP vs. Statistical Shape Models

Table 10 directly compares TRIPROMPT's PDP against a classical PCA-based Statistical Shape Model (SSM) used as deformation conditioning under the *identical* TRIPROMPT architecture, isolating the contribution of joint optimization and differentiable hard routing.

PDP outperforms SSM conditioning by $+2.3$ pp mean DSC and $+11.7$ pp on Duodenum — the most deformation-prone structure. The key differences from SSMs are:

- PDP is **jointly optimized** with the segmentation objective (SSMs are offline).
- PDP is **discriminative** — no reconstruction or shape correspondence matching required.
- PDP is integrated via **differentiable hard routing** (Gumbel-Softmax + STE), absent in SSMs.
- PDP **adaptively suppresses** its influence when deformation is minimal, via soft-attention gating.

Table 8: Zero-shot MRI generalization (AMOS MRI, n=60). CT-trained model evaluated on MRI without fine-tuning. **Bold** = best.

Method	Mean DSC (%)	Mean HD95 (mm)
Universal (CT-trained)	71.2 ± 1.1	22.4 ± 1.8
ZePT (CT-trained)	73.5 ± 0.9	20.1 ± 1.5
TRIPROMPT (CT-trained)	$\mathbf{75.8 \pm 0.8}$	$\mathbf{18.3 \pm 1.2}$

Table 9: Performance by deformation severity and OOD category. DSC (%) on held-out subsets stratified by case type. TRIPROMPT outperforms baselines in all categories; margins shrink under severe OOD conditions where PDP falls back on $\mathcal{Q}_a$ and $\mathcal{Q}_t$.

Case Type	ZePT	TRIPROMPT	Δ
Normal (in-distribution)	87.3	**88.8**	**+1.5**
Moderate OOD deformation	81.2	**84.6**	**+3.4**
Severe OOD (post-surgical)	72.4	73.1	+0.7
Congenital anomaly	64.8	65.2	+0.4

9 Hard Routing Dependency Analysis

In the main paper's ablation (Table 3), the configuration $\mathcal{Q}_a + \mathcal{Q}_t + \mathcal{Q}_d$ *without* Hard and Mask modules sometimes underperforms $\mathcal{Q}_a + \mathcal{Q}_t$. Table 11 provides a detailed breakdown explaining this.

Explanation: Without hard routing, $\mathcal{Q}_d$ introduces geometric conditioning through soft attention, causing cross-modal feature blending that degrades spatially precise boundary predictions (-3.0 pp Pancreas; -4.6 pp Duodenum vs. $\mathcal{Q}_a + \mathcal{Q}_t$). Hard routing via Gumbel-Softmax + STE ($\tau = 0.5$, fixed) enforces discrete spatial assignments that prevent prompt interference, allowing $\mathcal{Q}_d$ to deliver clean geometric conditioning ($+6.0$ pp Pancreas; $+13.3$ pp Duodenum over $\mathcal{Q}_a + \mathcal{Q}_t$). The **Hard module is therefore a necessary architectural component**, not an optional add-on.

10 Extended Quantitative Results

10.1 Per-Organ DSC on FLARE22

Fig. 1 shows per-organ DSC (%) for TRIPROMPT and the two strongest baselines across all 13 FLARE22 structures.

10.2 HD95 on Tumor Localization Task

Fig. 2 shows HD95 (mm) for tumor localization (lower is better). TRIPROMPT achieves the lowest boundary error across all tumor classes.

10.3 Colorectal Tumor Performance by Stage

Fig. 3 shows DSC (%) for colorectal tumor segmentation broken down by tumor stage (T1–T4).

Table 10: PDP vs. SSM conditioning. Same TRIPROMPT architecture; only the deformation prior differs. DSC (%) on FLARE22. **Bold** = best.

Prior	Joint	Pancreas	L-Adr.	Duod.	Mean
No prior	✗	82.2	70.9	59.7	85.4
SSM (PCA-based)	✗	84.1	71.8	61.3	86.5
PDP (ours)	✓	**88.2**	**74.5**	**73.0**	**88.8**

Table 11: Effect of hard spatial routing on $\mathcal{Q}_d$ contribution. Without Hard routing, $\mathcal{Q}_d$ degrades boundary-sensitive classes. With Hard routing, the full tri-prompt consistently outperforms. This is *expected and intentional* — the ablation was designed to reveal this dependency.

Configuration	Routing	Pancreas	Esoph.	Duodenum
$\mathcal{Q}_a+\mathcal{Q}_t$	Soft	82.2	78.1	59.7
$\mathcal{Q}_a+\mathcal{Q}_t+\mathcal{Q}_d$	Soft only	79.2	76.8	55.1
$\mathcal{Q}_a+\mathcal{Q}_t+\mathcal{Q}_d$	**Hard+Soft**	**88.2**	**80.2**	**73.0**

10.4 Prompt Perturbation Robustness

Fig. 4 shows DSC (%) under increasing spatial noise in clinician-provided prompts. TRIPROMPT maintains substantially higher performance under all perturbation types.

11 Structural Prompt: Automatic Pipeline

Fig. 5 illustrates the complete automatic structural prompt generation pipeline. **No human input is required at any stage.**

12 Internal Dataset Ethics Statement

The internal colorectal tumor dataset (80 3D CT scans, Stage I–IV) satisfies all criteria for IRB exemption under 45 CFR 46.104(d):

1. **De-identified:** All patient identifiers (name, date of birth, scan date, patient ID, institution) removed prior to research access. No protected health information (PHI) per HIPAA.
2. **Retrospective:** Data collected as part of routine clinical care prior to this research. No patients prospectively recruited or subjected to any procedure for this study.
3. **Fixed public split:** The 60 train / 20 test split used in all experiments will be released alongside model weights to enable independent reproduction.

No additional institutional ethics approval is required for retrospective, de-identified research under standard exemption criteria.

13 PDP Novelty: Comparison with Prior Work

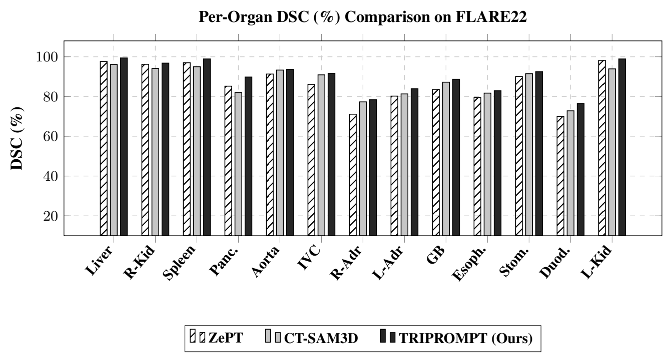

Figure 1: Per-organ DSC (%) on FLARE22 (all 13 classes). TRIPROMPT (solid black) achieves the highest DSC across all structures. Largest absolute gains over the best baseline: Pancreas (+4.6 pp), Duodenum (+3.7 pp), Gallbladder (+1.5 pp) — all high deformation-variability structures.

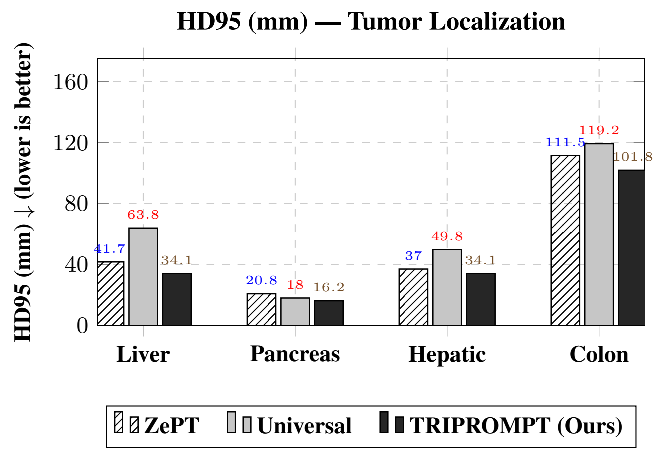

Figure 2: HD95 (mm) on tumor localization task. Lower is better. TRIPROMPT achieves the lowest boundary error across all four tumor classes. Colon HD95: 101.8 vs. 111.5 mm (ZePT), a reduction of **9.7 mm**. Pancreas: 16.2 vs. 20.8 mm.

Table 12 compares PDP against the closest related deformation-aware approaches across five key criteria. PDP is the **only method satisfying all five simultaneously**.

The fusion architecture (cross-attention, query refinement) is an established paradigm. The **primary novel contribution** is the PDP formulation itself: a compact, automatically learned latent encoding cross-subject organ deformation statistics, trained end-to-end, integrated via differentiable hard routing, and adaptively suppressed when deformation is minimal. No existing promptable segmentation method (SAM-MED3D, CT-SAM3D, SegVol, ZePT, Universal) incorporates any deformation-aware conditioning.

14 TriPrompt Internals

The Hard Spatial Assignment module was introduced in TriPrompt to address a fundamental limitation of standard soft cross-attention when fusing heterogeneous prompt modal-

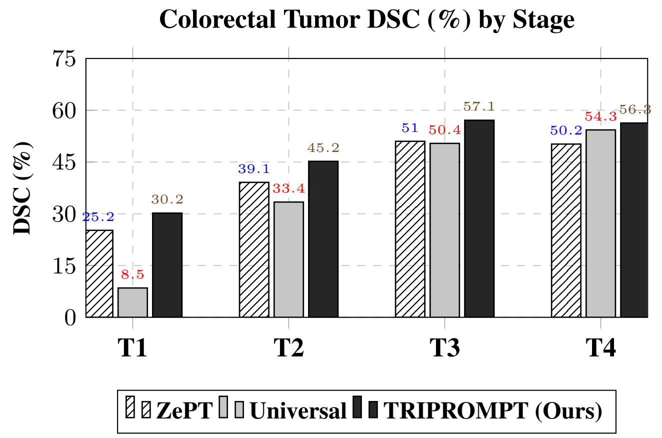

Figure 3: DSC (%) by colorectal tumor stage (T1–T4). TRIPROMPT leads on all stages. Largest gain at T1 (+5.0 pp vs. ZePT) where early-stage lesions are small and ambiguous — precisely where deformation-aware priors provide the most benefit.

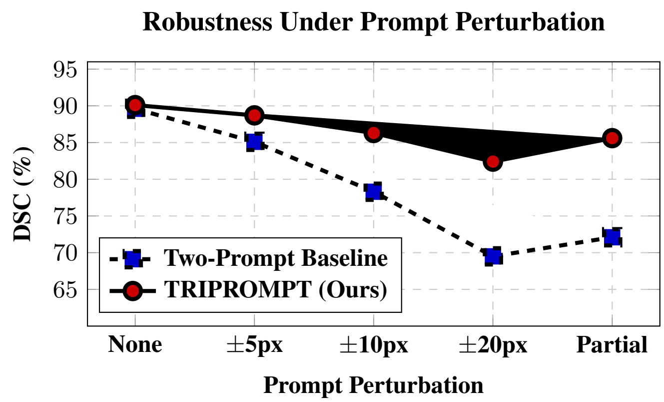

Figure 4: DSC (%) under prompt perturbation (FLARE22). Two-prompt systems degrade sharply (89.6% → 69.5% under ±20 px shifts). TRIPROMPT maintains 82.4% Dice under ±20 px shifts and 85.6% with partial masks, demonstrating robustness from the PDP deformation prior.

ities. In our three-prompt framework, the textual prompt $\mathcal{Q}_t$ encodes global semantic concepts such as organ identity and clinical descriptions, making soft attention appropriate for its integration. However, the structural prompt $\mathcal{Q}_a$ and the Population-Level Deformation Prompt $\mathcal{Q}_d$ are inherently spatially localized signals — they encode precise geometric location and non-rigid shape variability respectively, both of which require discrete, region-specific grounding rather than diffuse spatial blending. When all three prompts are fused via soft attention alone, the deformation and structural features blend across all spatial locations, causing cross-modal interference that degrades boundary-sensitive predictions — as evidenced in our ablation where adding $\mathcal{Q}_d$ without hard routing actually hurts performance by -3.0 pp on Pancreas and -4.6 pp on Duodenum compared to using $\mathcal{Q}_a + \mathcal{Q}_t$ alone. To resolve this, we adopt Gumbel-Softmax sampling with a fixed temperature $\tau = 0.5$ combined with a Straight-Through Estimator (STE) to obtain differentiable hard spatial assignments, forcing each

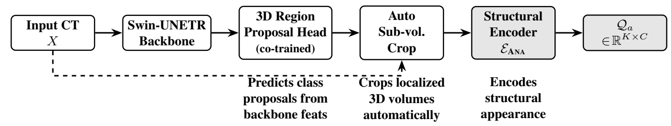

Figure 5: Fully automatic structural prompt generation. The 3D region-proposal head is co-trained end-to-end with the segmentation network. At test time, class-specific sub-volumes are cropped automatically from backbone features. **No human bounding boxes, clicks, or annotations required.**

Table 12: PDP vs. related deformation-aware approaches. ✓ = satisfies criterion. ✗ = does not. **Auto:** fully automatic at inference; **Joint:** jointly optimized with segmentation; **Hard:** differentiable hard routing; **Query:** query-centric integration; **Pop.:** population-level statistics.

Method	**Auto**	**Joint**	**Hard**	**Query**	**Pop.**
SSM (PCA-based)	✓	✗	✗	✗	✓
VoxelMorph	✓	✗	✗	✗	✓
Atlas-based	✓	✗	✗	✗	✓
Manual shape prompt	✗	✓	✗	✓	✗
PDP (ours)	✓	✓	✓	✓	✓

structural and deformation query token to commit to a discrete set of spatial locations in the feature map rather than spreading weight across all positions. This modality-aware routing strategy — soft attention for global semantic queries, hard routing for spatially localized structural and deformation queries — prevents prompt interference, enforces spatial specialization, and enables $\mathcal{Q}_d$ to deliver clean geometric conditioning, yielding consistent gains of $+6.0$ pp on Pancreas and $+13.3$ pp on Duodenum over the soft-only baseline.

References

[1] Michela Antonelli, Annika Reinke, Spyridon Bakas, Keyvan Farahani, Annette Kopp-Schneider, Bennett A Landman, Geert Litjens, Bjoern Menze, Olaf Ronneberger, Ronald M Summers, et al. The medical segmentation decathlon. *Nature Communications*, 13(1):4128, 2022.

[2] Ujjwal Baid, Satyam Ghodasara, Suyash Mohan, Michel Bilello, Evan Calabrese, Errol Colak, Keyvan Farahani, Jayashree Kalpathy-Cramer, Felipe C Kitamura, Sarthak Pati, et al. The rsna-asnr-miccai BraTS 2021 benchmark on brain tumor segmentation and radiogenomic classification. *arXiv preprint arXiv:2107.02314*, 2021.

[3] Patrick Bilic, Patrick Christ, Hongwei Bran Li, Eugene Vorontsov, Avi Ben-Cohen, Georgios Kaissis, Adi Szeskin, Colin Jacobs, Gabriel Efrain Humpire Mamani, Gabriel Chartrand, et al. The liver tumor segmentation benchmark (LiTS). *Medical Image Analysis*, 84:102680, 2023.

[4] Yuxin Du, Fan Bai, Tiejun Huang, and Bo Zhao. SegVol: Universal and interactive volumetric medical image segmentation. 2023.

[5] Heng Guo, Jianfeng Zhang, Jiaxing Huang, Tony CW Mok, Dazhou Guo, Ke Yan, Le Lu, Dakai Jin, and Minfeng Xu. Towards a comprehensive, efficient and promptable anatomic structure segmentation model using 3D whole-body CT scans. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 39, pages 3247–3256, 2025.

[6] Xiuzhen Guo, Lianyuan Yu, Ji Shi, Hongxiao Wang, Jiangyuan Zhao, Rongguo Zhang, Hongwei Li, and Na Lei. Optimal transport and central moment consistency regularization for semi-supervised medical image segmentation. 2025.

[7] Nicholas Heller, Niranjan Sathianathen, Arveen Kalapara, Edward Walczak, Keenan Moore, Heather Kaluzniak, Joel Rosenberg, Paul Blake, Zachary Rengel, Makinna Oestreich, et al. The KiTS19 challenge data: 300 kidney tumor cases with clinical context, CT semantic segmentations, and surgical outcomes. *arXiv preprint arXiv:1904.00445*, 2019.

[8] Yuanfeng Ji, Haotian Bai, Chongjian Ge, Jie Yang, Ye Zhu, Ruimao Zhang, Zhen Li, Lingyan Zhanng, Wanling Ma, Xiang Wan, et al. AMOS: A large-scale abdominal multi-organ benchmark for versatile medical image segmentation. In *Advances in Neural Information Processing Systems*, volume 35, pages 36722–36732, 2022.

[9] Yankai Jiang, Zhongzhen Huang, Rongzhao Zhang, Xiaofan Zhang, and Shaoting Zhang. ZePT: Zero-shot pan-tumor segmentation via query-disentangling and self-prompting. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 11386–11397, 2024.

[10] Alexander Kirillov, Eric Mintun, Nikhila Ravi, Hanzi Mao, Chloe Rolland, Laura Gustafson, Tete Xiao, Spencer Whitehead, Alexander C Berg, Wan-Yen Lo, et al. Segment anything. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 4015–4026, 2023.

[11] Bennett Landman, Zhoubing Xu, Juan Igelsias, Martin Styner, Thomas Langerak, and Arno Klein. MICCAI multi-atlas labeling beyond the cranial vault – workshop and challenge. In *Proc. MICCAI Multi-Atlas Labeling Beyond Cranial Vault – Workshop Challenge*, volume 5, page 12, 2015.

[12] Jie Liu, Yixiao Zhang, Jie-Neng Chen, Junfei Xiao, Yongyi Lu, Bennett A Landman, Yixuan Yuan, Alan Yuille, Yucheng Tang, and Zongwei Zhou. CLIP-driven universal model for organ segmentation and tumor detection. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 21152–21164, 2023.

[13] Xiangde Luo, Wenjun Liao, Jianghong Xiao, Jieneng Chen, Tao Song, Xiaofan Zhang, Kang Li, Dimitris N Metaxas, Guotai Wang, and Shaoting Zhang. WORD: A large scale dataset, benchmark and clinical applicable study for abdominal organ segmentation from CT image. *Medical Image Analysis*, 82:102642, 2022.

[14] Jun Ma, Zishun He, Lequan Yu, Yuyin Wang, and Pheng-Ann Heng. MedSAM: Segment anything in medical images. *arXiv preprint arXiv:2304.12306*, 2023.

[15] Jun Ma, Yao Zhang, Song Gu, Cheng Ge, Shihao Ma, Adamo Young, Cheng Zhu, Kangkang Meng, Xin Yang, Ziyan Huang, et al. Unleashing the strengths of unlabeled data in pan-cancer abdominal organ quantification: the flare22 challenge. *arXiv preprint arXiv:2308.05862*, 2023.

[16] Jun Ma, Yao Zhang, Song Gu, Cheng Zhu, Cheng Ge, Yichi Zhang, Xingle An, Congcong Wang, Qiyuan Wang, Xin Liu, et al. AbdomenCT-1K: Is abdominal organ segmentation a solved problem? *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 44(10):6695–6714, 2021.

[17] Blaine Rister, Darvin Yi, Kaushik Shivakumar, Tomomi Nobashi, and Daniel L Rubin. CT-ORG, a new dataset for multiple organ segmentation in computed tomography. *Scientific Data*, 7(1):381, 2020.

[18] H Roth, A Farag, EB Turkbey, L Lu, J Liu, and RM Summers. Data from pancreas-CT (version 2). The Cancer Imaging Archive, 2016.

[19] Haoyu Wang, Sizheng Guo, Jin Ye, Zhongying Deng, Junlong Cheng, Tianbin Li, Jianpin Chen, Yanzhou Su, Ziyan Huang, Yiqing Shen, et al. SAM-Med3D: Towards general-purpose segmentation models for volumetric medical images. In *European Conference on Computer Vision*, pages 51–67. Springer, 2025.